

VIHTI MAASOJA, Finland

WMO: 027580

Lat: 60.419N

Lon: 24.399E

Elev: 44

StdP: 100.80

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-21.6	-28.4	0.3	-26.0	-23.6	0.4	-21.4	7.9	-2.8	6.8	-2.4	0.0	180	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	27.7	19.4	25.7	18.2	23.9	17.0	20.9	25.7	19.5	24.1	18.2	22.3	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.0	23.0	17.8	12.9	21.5	16.6	11.9	20.1	60.4	25.8	55.8	24.1	51.7	22.3	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.7	5.0	DB	-27.1	29.4	6.4	2.4	-31.7	31.1	-35.4	32.5	-39.0	33.9	-43.7	35.6	(4)
(5)			WB	-26.6	22.0	6.4	1.4	-31.3	23.1	-35.0	23.9	-38.6	24.7	-43.3	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-6.8	-5.0	-2.3	3.7	9.9	13.6	16.5	14.8	10.7	4.7	1.5	-2.8	(6)	
	DBStd	9.03	6.97	6.68	5.14	3.36	4.10	3.28	3.15	2.88	3.23	3.61	4.42	5.78	(7)	
	HDD10.0	2423	521	421	381	191	53	7	0	2	31	167	255	396	(8)	
	HDD18.3	4922	779	654	639	440	263	146	76	114	231	422	505	654	(9)	
	CDD10.0	572	0	0	0	1	49	114	202	152	50	3	0	0	(10)	
	CDD18.3	28	0	0	0	0	1	3	19	6	0	0	0	0	(11)	
	CDH23.3	508	0	0	0	0	42	93	283	89	1	0	0	0	(12)	
	CDH26.7	94	0	0	0	0	4	11	62	17	0	0	0	0	(13)	
	WSAvg	2.0	1.8	2.0	2.2	2.0	2.2	2.3	1.9	1.8	1.8	2.0	2.0	2.0	(14)	
(15) Precipitation	PrecAvg	694	64	43	37	33	40	65	74	79	53	83	74	58	(15)	
	PrecMax	843	109	106	66	58	89	111	202	168	130	195	173	103	(16)	
	PrecMin	534	11	2	6	6	8	23	1	9	9	31	11	8	(17)	
	PrecStd	86	24	27	16	16	23	27	48	37	33	38	39	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	5.7	10.6	19.6	27.1	28.2	30.4	28.9	22.1	14.3	10.5	8.6	(19)	
		MCWB	3.7	3.2	5.8	10.8	16.3	18.3	20.6	19.6	17.0	12.2	9.7	8.1	(20)	
	2%	DB	3.5	3.9	7.6	16.8	24.2	26.0	28.4	25.6	19.8	12.7	9.2	6.5	(21)	
		MCWB	3.0	3.0	4.2	9.2	14.9	17.2	20.4	18.6	15.8	11.3	8.8	5.8	(22)	
	5%	DB	2.4	2.8	5.4	13.9	22.0	23.9	26.5	23.6	18.2	11.8	7.9	5.0	(23)	
		MCWB	2.1	2.2	3.0	8.0	13.8	16.4	19.4	17.3	14.5	10.6	7.4	4.5	(24)	
	10%	DB	1.3	1.9	3.8	11.2	19.5	21.6	24.5	21.7	16.8	10.7	7.0	3.6	(25)	
		MCWB	1.0	1.5	2.2	6.3	12.6	15.1	18.4	16.4	14.3	9.7	6.5	3.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	4.0	6.2	11.6	18.0	20.5	22.4	21.9	17.7	13.2	10.0	8.3	(27)	
		MCDB	4.2	4.8	8.8	18.1	23.9	25.5	27.0	26.0	20.5	13.9	10.4	8.5	(28)	
	2%	WB	3.1	3.1	4.6	10.0	16.2	18.7	21.3	19.7	16.6	12.0	8.9	6.1	(29)	
		MCDB	3.5	3.8	6.9	16.0	21.3	23.9	26.2	23.5	18.9	12.5	9.3	6.5	(30)	
	5%	WB	2.2	2.3	3.5	8.4	14.9	17.3	20.4	18.4	15.6	11.0	7.6	4.6	(31)	
		MCDB	2.5	2.8	5.1	12.7	20.9	22.3	25.3	22.0	17.4	11.5	7.9	5.1	(32)	
	10%	WB	1.1	1.5	2.5	7.0	13.2	16.0	19.1	17.4	14.6	9.8	6.6	3.2	(33)	
		MCDB	1.3	1.9	3.5	10.5	18.4	20.5	23.7	20.5	16.2	10.5	7.0	3.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.8	9.1	10.9	13.0	11.6	12.0	11.9	10.0	7.4	4.5	5.0	(35)	
		MCDBR	3.9	5.3	9.8	17.1	18.1	16.4	15.3	15.6	12.5	8.8	5.0	5.0	(36)	
		MCWBR	3.7	4.5	6.6	9.9	9.3	8.5	8.1	9.1	8.4	7.2	4.7	4.7	(37)	
	5% WB	MCDBR	3.8	4.8	8.3	14.8	15.4	13.4	13.8	12.4	10.7	6.8	5.0	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.272	0.309	0.332	0.361	0.358	0.355	0.378	0.370	0.342	0.331	0.300	N/A	(40)	
		taud	2.125	2.189	2.258	2.357	2.420	2.459	2.415	2.435	2.506	2.398	2.213	N/A	(41)	
	Edn at Noon		471	670	788	836	868	876	842	820	785	654	436	N/A	(42)	
			42	72	92	101	102	100	102	93	73	59	39	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.25	0.81	2.30	3.84	5.20	5.40	5.21	4.03	2.46	1.07	0.30	0.14	(44)	
	RadStd		0.04	0.17	0.29	0.40	0.56	0.43	0.48	0.42	0.32	0.17	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	-138	N/A	N/A	N/A	

Nomenclature: See separate page